

TOKUSHIMA, Japan

WMO: 478950

Lat: 34.0678N

Lon: 134.5733E

Elev: 21

StdP: 14.68

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.8	34.5	16.6	12.7	39.6	19.0	14.3	40.5	20.9	43.7	19.0	43.1	7.1	290	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.9	92.2	78.1	90.4	77.8	88.7	77.3	80.1	88.0	79.3	86.9	78.5	85.8	9.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.2	146.4	84.3	77.2	141.8	83.7	76.3	137.6	83.2	43.9	88.1	42.9	87.0	42.0	86.0	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.7	17.1	15.1	DB	29.5	96.2	2.0	1.7	28.0	97.4	26.8	98.4	25.7	99.4	24.2	100.6
			WB	25.6	81.4	2.2	1.0	24.1	82.1	22.8	82.7	21.6	83.2	20.0	83.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.7	43.3	44.3	49.9	59.0	67.6	73.7	81.0	83.4	77.1	67.3	56.9	47.7
	DBStd	14.52	3.95	4.68	5.33	5.32	3.83	3.76	3.69	2.99	4.46	4.81	5.26	4.85
	HDD50	563	212	171	70	3	0	0	0	0	0	0	5	102
	HDD65	2740	674	580	469	187	18	0	0	0	0	29	247	536
	CDD50	5202	3	10	66	275	546	712	962	1035	813	537	212	31
	CDD65	1905	0	0	0	8	99	262	497	570	363	102	4	0
	CDH74	16393	0	0	0	23	313	1285	4994	6625	2843	309	1	0
	CDH80	5419	0	0	0	1	23	205	1724	2696	743	27	0	0
Wind	WSAvg	7.0	7.6	7.5	7.6	7.5	7.3	6.4	6.6	7.0	6.8	6.5	6.4	7.2
Precipitation	PrecAvg	65.2	1.9	2.6	3.9	4.3	5.9	8.0	6.9	7.4	10.7	6.8	4.1	2.7
	PrecMax	98.2	4.4	5.8	7.0	8.9	17.0	15.0	15.9	37.5	33.4	18.5	11.1	9.8
	PrecMin	39.1	0.3	0.6	1.8	1.6	1.8	2.2	2.5	0.9	2.7	0.7	0.6	0.2
	PrecStd	16.1	1.1	1.4	1.5	1.9	3.7	3.0	3.3	7.7	6.9	4.4	2.8	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.5	64.4	69.0	77.1	82.7	88.9	94.2	95.7	91.1	83.2	72.7	64.7
		MCWB	51.3	57.5	57.7	62.7	64.5	75.6	77.8	77.8	76.9	72.8	63.1	57.7
	2%	DB	54.9	58.9	65.4	73.1	79.7	84.4	91.4	93.0	88.5	79.5	70.1	60.7
		MCWB	47.7	51.8	56.2	60.3	64.2	72.6	78.3	78.3	75.8	69.8	61.9	53.8
	5%	DB	52.6	55.3	62.5	70.5	77.5	82.2	89.6	91.3	86.4	77.3	68.0	58.0
		MCWB	45.2	48.2	53.8	59.9	64.3	71.3	77.8	78.0	74.4	68.0	60.4	50.7
	10%	DB	50.4	52.5	59.7	68.1	75.3	80.2	87.8	89.7	84.3	75.2	65.8	55.7
		MCWB	43.2	45.4	52.0	58.1	64.0	70.4	77.3	77.7	73.9	66.3	58.7	48.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.0	58.8	62.0	67.5	72.4	77.6	81.0	81.7	79.7	76.1	67.1	60.2
		MCDB	57.1	63.4	65.8	72.2	76.7	84.8	88.9	89.7	86.1	80.5	69.9	62.2
	2%	WB	49.4	53.5	58.4	64.5	69.9	75.8	79.9	80.6	78.3	72.9	64.3	55.2
		MCDB	53.3	58.1	62.9	69.2	74.0	81.4	87.7	88.7	84.3	77.4	68.0	59.7
	5%	WB	46.3	49.8	55.7	62.2	67.9	74.5	79.0	79.8	77.3	70.2	61.9	52.1
		MCDB	51.3	54.0	60.9	67.8	72.8	79.5	86.5	87.7	83.1	74.9	66.7	56.8
	10%	WB	44.0	46.4	53.0	60.3	66.3	73.2	78.2	79.0	76.2	67.9	59.6	49.4
		MCDB	49.3	51.3	58.9	66.0	72.0	77.7	85.3	86.7	82.0	73.6	64.8	54.8
Mean Daily Temperature Range	5% DB	MDBR	11.4	12.4	14.0	14.6	13.9	11.2	10.8	11.9	11.2	11.7	12.2	11.8
		MCDBR	14.6	16.4	18.1	17.5	17.2	13.9	13.2	14.0	13.0	13.4	14.0	14.7
		MCWBR	10.2	11.9	12.6	10.2	8.7	6.2	5.2	5.6	5.9	7.4	9.1	10.3
	5% WB	MCDBR	14.1	15.3	16.6	14.5	12.7	10.9	12.0	12.5	10.8	11.1	12.5	14.1
		MCWBR	11.5	13.2	13.8	10.4	8.6	5.7	5.2	5.6	5.7	7.5	9.8	11.8
Clear-Sky Solar Irradiance	taub	0.400	0.448	0.509	0.542	0.548	0.563	0.562	0.534	0.497	0.459	0.424	0.394	
	taud	2.170	2.051	1.905	1.878	1.934	1.969	2.019	2.090	2.161	2.185	2.188	2.203	
	Ebn at Noon	248	248	243	242	242	237	237	241	244	242	238	242	
	Edh at Noon	37	47	59	63	60	58	55	50	45	41	36	34	
All-Sky Solar Radiation	RadAvg	817	1002	1306	1545	1648	1461	1625	1682	1305	1066	831	733	
	RadStd	58	77	91	152	163	150	197	152	114	110	72	47	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page